

Uncertainty Analysis of a Strain-Instrumented Cantilever Beam Scale for Bottle Weight and Gulp-Size Measurement

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Section 11690 February 28, 2026

Abstract—This study evaluated a strain-instrumented cantilever beam as a low-cost scale for estimating bottle weight and average gulp size using two independent methods: a calibration-based voltage-to-weight relationship and a Mechanics of Materials (MoM) model based on beam bending and strain. A 6061-T6 aluminum beam fitted with a bonded axial strain gauge was connected to a quarter Wheatstone bridge, amplifier, and data acquisition system, and each recorded data point was formed from 2500 voltage samples. Do not cite references in the abstract. An abstract should have One unloaded point, and four applied loads were used for calibration. Monte Carlo simulation was used to determine the uncertainty in the calibration constant, and root-sum-square techniques were used to propagate uncertainty in the final calculated numbers. Although both approaches recorded the anticipated drop in bottle weight during the drinking sequence, the calibration method yielded more precise estimates overall. In comparison to a laboratory scale measurement of 2.4444 N, the calibration approach yielded 2.4408 N for the entire bottle, whereas the MoM method yielded 2.0536 N. For the empty bottle, the calibration and MoM methods gave 0.0708 N and 0.0596 N, respectively, compared with a scale measurement of 0.0789 N. The average gulp size was 0.395 N using the calibration method and 0.332 N using the MoM method. These results show that a strain-gauge-based cantilever beam can provide reliable weight estimates when combined with calibration and uncertainty analysis.

Index Terms— Calibration, Cantilever Beam, Data Acquisition, Strain Gauge, Uncertainty Propagation, Wheatstone Bridge.

I. INTRODUCTION

THE goal of this experiment is to create a strain-instrumented cantilever beam scale and assess its capacity to estimate average gulp weight and beverage-can weight using both an empirical voltage-to-weight calibration and an analytical Mechanics of Materials (MoM) model. A 6061-T6 aluminum beam clamped as a cantilever and a bonded axial strain gauge placed close to the fixed end make up the scale. A data acquisition device (DAQ) running a LabVIEW virtual instrument (VI) records the amplified bridge output after the strain gauge is linked to a quarter Wheatstone strain bridge (WSB) installed on an amplifier printed circuit board (PCB). Each saved data point was formed from 2500 voltage samples, and the LabVIEW virtual instrument (VI) recorded the

corresponding sample standard deviation for that set. [8] Two independent approaches are used to estimate applied load: a Mechanics of Materials (MoM) method based on beam-bending theory and measured strain, and a calibration method that maps measured voltage directly to known applied weights. The preferred method is selected by comparing agreement and expanded uncertainty at the 95% confidence level, with particular attention to how amplification, baseline removal (taring), and DAQ quantization influence small-signal measurements. [1], [2], [3].

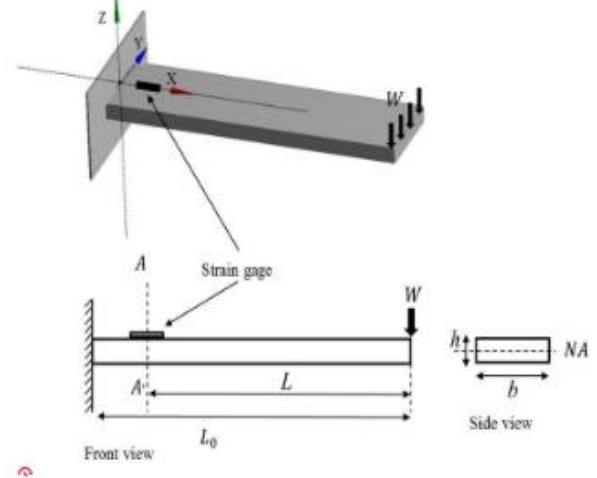


Fig 1. Cantilever beam scale configuration showing the fixed support, axial strain gauge location near the clamped end, applied end load.

Fig. 1 defines the cantilever geometry and parameters utilized in (1)– (7). The bending moment is determined by applying a concentrated load W (N) at an effective lever arm distance L (m) from the strain gauge position.

$$M = WL \quad (1)$$

where bending moment ($\text{N}\cdot\text{m}$) is represented by M . The flexure relation controls the equivalent normal surface stress under bending.

$$\sigma_x = \frac{Mc}{I} \quad (2)$$

where σ_x is bending stress (Pa), c is the distance from the neutral axis to the outer fiber (m), and I is the second moment of area (m^4). For a rectangular beam cross section,

$$I = \frac{bh^3}{12} \quad (3)$$

where b is beam width (m) and h is beam thickness (m), and the outer-fiber distance is.

$$c = \frac{h}{2} \quad (4)$$

Assuming linear elastic behavior, stress and strain are related by Hooke's law,

$$\varepsilon = \frac{\sigma_x}{E} \quad (5)$$

where E is Young's modulus (Pa) and ε is normal strain (unitless). The yield strength $\sigma_y \approx 276$ MPa and the typical published characteristics for 6061-T6 list $E \approx 68.9$ GPa, which serve as benchmarks for assessing elastic performance. [7]

Combining (1)–(5) and solving for load yields the MoM load estimate,

$$W_{MoM} = \frac{Eb h^2 \varepsilon}{6L} \quad (6)$$

Enforcing yielding in the flexure structure allows for the estimation of a conservative elastic operating bound, preventing irreversible deformation and maintaining repeatability.

$$W_{max} = \frac{\sigma_y b h^2}{6L} \quad (7)$$

where σ_y is the material yield strength (Pa). [7]

The bonded axial strain gage used in this experiment has a nominal resistance of $R_0 = 120 \Omega$, so elastic loading produces only a small fractional change $\Delta R/R_0$ that must be converted to a measurable voltage. [1]

The strain gage converts mechanical strain into a small resistance change. The nominal resistance of a conductor can be expressed as

$$R = \frac{\rho l}{A} \quad (8)$$

where R is resistance (Ω), ρ is electrical resistivity ($\Omega \cdot \text{m}$), l is conductor length (m), and A is conductor cross-sectional area (m^2). Strain is related to fractional resistance change through the gage factor G_f (unitless),

$$\varepsilon = \frac{\Delta R}{R G_f} \quad (9)$$

where ΔR is the resistance change (Ω). The course handout specifies the gage factor and tolerance for the gages used in this lab, and these values are carried into the uncertainty model. [1] [5]

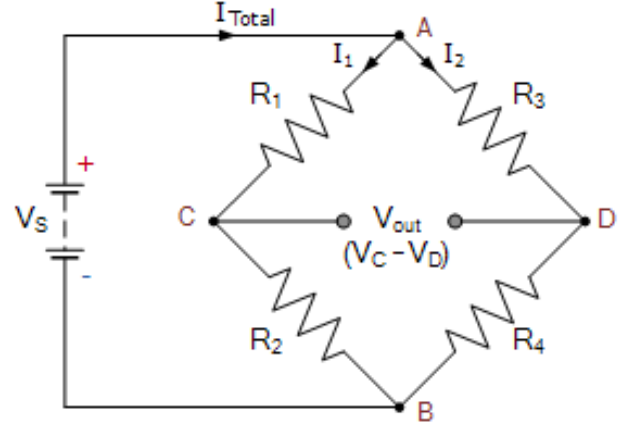


Fig 2. excitation V_s , bridge output ΔV_g , amplifier output V_{amp} , and DAQ connections.

In this setup, V_s is the bridge excitation, V_g is the unamplified bridge output used during bridge balancing/centering, and V_{amp} is the amplified output recorded for analysis; after taring, voltage changes are reported as ΔV_g and ΔV_{amp} . [1], [4]

The WSB/amplifier configuration and voltage definitions used in (10)–(13) are shown in Fig. 2. For a quarter Wheatstone bridge under small-change assumptions, strain is related to bridge output by.

$$\varepsilon = \frac{4\Delta V_g}{V_s G_f} \quad (10)$$

where ΔV_g is bridge differential output (V) and V_s is excitation of bridges (V). The bridge/amplifier must be powered by the DAQ's 3.3 V supply. Before taring, the unloaded output must be balanced or centered close to 0 V using onboard potentiometer trim adjustment. The lab approach eliminates baseline offsets so that future measurements reflect load-induced changes rather than beginning bias because baseline offsets are predicted due to installation strain, adhesive effects, and wiring imbalance [1], [4]. [2], [3].

To improve sensitivity prior to digitization, the bridge output is amplified. The amplified change is related to the bridge differential change by

$$\Delta V_{amp} = G \Delta V_g \quad (11)$$

where ΔV_{amp} is amplifier gain (unitless) and is amplified voltage change (V). [1] When (11) is substituted into (10) strain is produced in terms of enhanced output.

$$\varepsilon = \frac{4\Delta V_{amp}}{GV_s G_f} \quad (12)$$

Baseline offsets are removed by defining a tared amplified signal,

$$\Delta V_{amp} = V_{amp} - V_{tare} \quad (13)$$

where V_{amp} is the amplifier output (V) that is measured, and V_{tare} is the baseline output (V).

V_{amp} is recorded by the DAQ as an analog voltage and uses an analog-to-digital converter (ADC) to transform it into digital values. ADC bit depth and the chosen input range (sample window). 14-bit ± 10 V ADC inputs with adjustable gain and sample rate are provided by the SADI DAQ utilized in this course. [6] The chosen input window is quantified using a 14-bit ADC also known as the least significant bit (LSB). The size of the voltage step is.

$$Resolution = \frac{V_{max} - V_{min}}{2^n - 1} \quad (14)$$

where n is the ADC bit depth (bits) and V_{max} and V_{min} are the sample-window boundaries (V). The technique gives the following sample windows: ± 0.64 V, ± 1.28 V, ± 2.56 V, ± 5.12 V, and ± 10.24 V. The smallest window that remained unsaturated under the maximum predicted loading was selected using a saturation check. [2] In the Results and Discussion sections, the lab discussion focuses on reporting the ADC/sample-window parameters and relating those decisions to uncertainty and saturation behavior. [3]

An end-to-end MoM estimate in terms of the recorded electrical signal is obtained by combining the mechanical model with the strain-voltage relationship. Substituting (12) into (6) yields

$$W_{MoM} = \frac{4\Delta V_{amp}}{GV_s G_f} \cdot \frac{Ebh^2}{6L} \quad (15)$$

By applying known masses and fitting a linear connection between applied load and measured voltage, a calibration method employs the instrumented beam as a transducer in parallel with the analytical approach. The calibrated estimate is expressed as

$$W_{cal} = c_{cal} \Delta V_{amp} \quad (16)$$

where c_{cal} is the calibration constant (N/V) and W_{cal} is the calibrated weight (N). In order to support analysis, the lab protocol calls for collecting repeated sample sets for each loading condition and employing a minimum of four calibration masses, ranging in weight from 20g to 200 g. [2] The course handout also makes a distinction between the tared signal, which is used to measure changes brought on by load, and the amplifier output, which is utilized for the calibration workflow. [1]

The strategy utilized for gulp measurements is chosen based on an evaluation of uncertainty for both approaches. Monte Carlo simulation (MCS) is used to quantify the uncertainty of c_{cal} since it is derived from a linearized fit of V_{amp} and calibration weights. Expanded uncertainties at the 95% confidence level are utilized to compare agreement and determine which approach yields the best accurate estimate for the can and gulp outcomes. Remaining uncertainties are propagated via the governing equations using a root-sum-square (RSS) framework. [3]

The smallest non-saturating sample window and amplification are expected to increase the signal magnitude in relation to the LSB indicated by (14), hence reducing quantization-driven uncertainty. Since such effects are absorbed into c_{cal} , the calibration approach should be less sensitive to uncertainty in E , b , h , and L . In contrast, the MoM method should react more strongly to beam geometry and material-property uncertainty. If elastic behavior is preserved and voltage saturation is prevented, careful balancing/taring and window selection should lead to improved agreement between approaches. [2], [3]

II. PROCEDURE

A. Strain Gage Mounting, Preparation, and Verification

The quarter-bridge mounting procedures were followed in order to prepare the beam surface at the designated gage point. Fine sandpaper was used to gently remove any paint or coatings, and denatured alcohol was used to wipe the surface of any debris and oils. To direct the strain gauge's axial alignment, alignment marks were polished onto the surface. The tape-hinge method was used to move the strain gage to the beam after it was set on a cleaned metal ruler. The gage was pressed into a tiny drop of cyanoacrylate adhesive at the hinge line, and the gage was bonded by applying firm pressure for about a minute. [9]

After removing about 1 cm of insulation from both ends, separating the conductors by about 3–4 cm at the circuit end, and tinning all exposed wire ends, a 45 cm length of two-conductor strain-gage wire was ready. Following the bonding of the gage, the clear tape was carefully removed, the gage-end conductors were bent downward, and a strain-relief "hump" was created 1-2 cm from the gage. Masking tape was used to attach the wire to the beam, covering the strain-relief hump on both sides. The wire and gage leads were soldered after being twisted. The installed strain gage, lead routing, and strain-relief arrangement are shown in Fig. 3. [9]

Using a multimeter, the installation was confirmed by making sure there was not a short and that the gage resistance was roughly the nominal amount. The gage nominal resistance for this lab's quarter-bridge completion setup was 120 Ω . [1], [9]

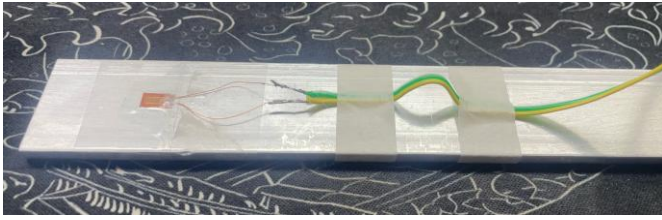


Fig 3. Bonded axial strain gage mounted near the fixed end of the beam, showing lead-wire attachment and strain-relief routing used prior to wiring the bridge/amplifier PCB.

B. Beam Mounting

The beam was installed as a cantilever in the lab station holder once the bottle placement reference and beam geometry were determined. Before tightening, the fixed end of the beam was completely set on the support face of the holder to reduce slip. In relation to the holder, the beam was positioned so that its long axis was level and straight. To secure the fixed end of the beam and stop motion during loading, the two Allen screws on the holder were then manually tightened. After tightening, the fixed end was visually inspected again to make sure there was no remaining space between the holder's face and the beam. After mounting, the bottle was briefly placed at the marked location to confirm that the load acted at the intended position and that the beam did not shift within the clamp. The overall beam-scale apparatus, including the cantilever beam, bottle placement location, WSB/amplifier PCB, and DAQ, is shown in Fig. 4.

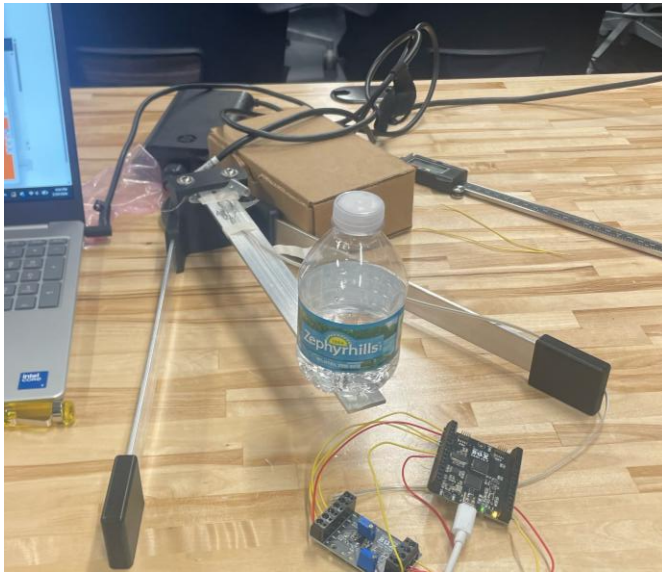


Fig 4. Experimental beam-scale setup showing the cantilever beam mounted in the holder.

C. WSB/Amplifier PCB Wiring and DAQ Connections

The quarter-bridge configuration was completed by connecting the gage leads to the Wheatstone strain bridge/amplifier after the strain gage was installed and the beam was fastened in the holder. By wiring DAQ +3.3 V $\rightarrow$ VIN and DAQ GND $\rightarrow$ GND in accordance with the course wiring diagram, the PCB was powered by the DAQ 3.3 V supply. [1], [4]

DAQ Channel 0 (ACH0) was connected to measure the bridge excitation voltage V_s by wiring the channel input across the bridge supply node and ground. DAQ analog input channels were then established to record the voltages needed by LabVIEW VI. By connecting the channel input to the amplifier output terminals listed in the course diagram, DAQ Channel 2 (ACH2) was set up to measure the amplifier output voltage V_{amp} . Since DAQ Channels 1 and 3 (ACH1 and ACH3) were not utilized in this lab, no data was recorded. The DAQ and WSB/amplifier PCB connections are shown in Fig. 5. [1], [4]

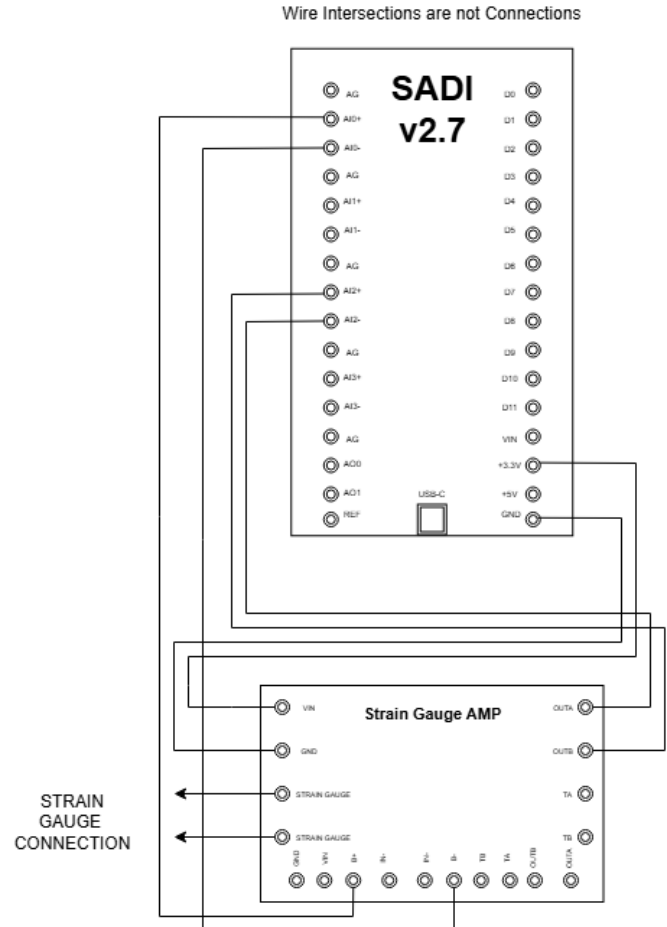


Fig 5. Wiring diagram of the DAQ and WSB/amplifier PCB connections used in this lab, including DAQ 3.3 V power, ground, bridge excitation.

D. LabVIEW Virtual Instrument Setup

DAQ signals were acquired and saved-point statistics were computed using the course LabVIEW framework. The VI calculated the mean and standard deviation over a 2500 sample frame while the DAQ collected continuous samples. These mean and standard-deviation values were saved as a single set of data points when the save button was pressed. [2], [8]

Therefore, the mean and standard deviation for ACH0 (V_s) and ACH2 (V_{amp}), calculated from the most recent 2500 samples, were included in each recorded point. The unloaded baseline V_{tare} was established by the VI tare function, and the tared amplified voltage ΔV_{amp} was used for future measurements. The VI also used the MoM and calibration procedures to calculate the corresponding weight for each saved

point. The recorded data and computed outputs were subsequently saved to a file for further examination. [2], [8]

E. Beam Geometry Measurements

A centerline mark was added on the beam to indicate the center of the water bottle, and the bottle was placed at the free end of the beam to control load placement. This centerline mark served as the placement reference for calibration and further measurements because the bottle was treated as a concentrated load operating via its center of gravity. After that, LabVIEW VI was used to measure and enter the beam geometry parameters needed by the MoM subVI [2].

A ruler was used to measure the lever arm distances L and L_0 . In this case, L_0 was the extra reference distance specified in the course diagram, and L was the distance between the strain gage site and the bottle centerline mark. Calipers were used to measure the beam thickness (h), and a micrometer was used to measure the beam width (b). To minimize reading error, each dimension was measured several times, and the average value was noted before the final numbers were entered into the VI. [1], [2]

To ensure uniform positioning for the ensuing saturation testing, calibration, repeatability trials, and gulp measurements, the bottle was lastly positioned with its center aligned with the designated centerline. [2]

F. Bridge Balancing, Amplifier Centering, and Taring

Before data collection, the Wheatstone strain bridge/amplifier (WSB/amp) printed circuit board (PCB) trim potentiometers were set to create a near-zero baseline with no load applied to the beam. The amplifier centering trim potentiometer was set to bring the unloaded amplifier output V_{amp} near 0 V. This modification was made after the desired channels and windows had been set up because altering the DAQ sample window required restarting the VI. [1], [2], [4]

The unloaded baseline V_{tare} was recorded using the LabVIEW tare function after the unloaded output had been centered. In order to reference load-induced voltage changes to the unloaded condition rather than the raw amplifier output, subsequent measurements were subsequently conducted using the tared amplified voltage ΔV_{amp} , as previously defined in the Introduction. Prior to calibration and each measurement set that needed a stable baseline, this balancing/centering and taring procedure was conducted. [1], [2], [4]

G. Sample Window Selection

The available DAQ sample windows were assessed to determine input ranges that would improve voltage resolution without causing saturation prior to recording calibration and weight measurements. This lab investigated sample windows of ± 0.64 V, ± 1.28 V, ± 2.56 V, ± 5.12 V, and ± 10.24 V [2].

To assess the baseline signal behavior for the observed channels, one saved point was collected at each available sample window without applying any load to the beam. Using these unloaded measurements, it was possible to determine whether the output was getting close to the window boundaries and whether the baseline signal stayed within the chosen range. For further testing and data collection, only sample windows that kept the unloaded signal within the selected range were

retained. [2]

The screening procedure was carried out independently for each channel since the amplified output V_{amp} and excitation voltage V_s were recorded on different DAQ channels. From this preliminary screening procedure, the final windows utilized for calibration, repeatability, and gulp measurements were chosen. [2], [3]

H. Calibration Data Collection

Four known weights weighing between 20 and 200 g were used for calibration, and before loading, an extra unloaded point of 0 g was recorded. The laboratory electronic scale was used to measure each calibration mass. Before fitting the calibration relationship, the recorded mass values were converted to reference weight in Newtons. [1], [2].

First, one unloaded saved point was acquired by mounting the beam, running the VI, and establishing the unloaded baseline using the previously stated tare technique. In order to ensure that the load acted through the same reference position used in the geometry measurements, each calibration mass was then positioned at the designated spot close to the beam's free end. A saved point was noted for that load once the beam response stabilized and the visual oscillations subsided. All four calibration masses followed this procedure. [2]

Excel was used to export the calibration data that had been saved. Using the observed amplifier output V_{amp} as the independent variable and reference weight from the laboratory scale as the dependent variable, a linear trendline was fitted. This best-fit line's slope was used as the calibration constant, or c_{cal} . This slope was then used with the tared amplifier signal ΔV_{amp} in the following weight calculations, in accordance with the course calibration instructions. Prior to the repeatability and gulp measurements, the resulting value of c_{cal} was entered into the VI. [1], [2]

As part of the uncertainty study, c_{cal} 's uncertainty was subsequently measured using Monte Carlo simulation as it was derived from a fitted trendline rather than a direct instrument reading. [8]

I. Bottle Placement Repeatability Trials

The system was first brought to an unloaded baseline utilizing the previously mentioned balance and tare technique in order to measure repeatability related to bottle placement on the beam. After that, the unopened water bottle was positioned close to the beam's free end so that its center lined up with the mark created by the bottle's centerline during the geometry measurements. A saved point was noted once the output stabilized and the oscillations were less noticeable. [2]

The bottle was then removed completely from the beam and placed again at the same marked location, and another saved point was recorded after the response stabilized. This remove-and-replace procedure was repeated for a total of 10 placement trials, with one saved point collected per placement. These 10 measurements were later treated as direct repeated measurements for statistical analysis, including the mean, standard deviation, and confidence interval associated with

placement repeatability. [2], [8]

J. Gulp Measurement Sequence

Following the completion of the placement repeatability experiments, the system was once more brought back to an unloaded baseline utilizing the previously mentioned tare technique. After the output stabilized and the visual oscillations stopped, the unopened water bottle was positioned at the designated spot on the beam's free end, and a saved point was noted. This was the first measurement of the entire bottle. [2]

After taking a gulp, the bottle was put back in its designated spot on the beam. Another saved point was made after the beam had steadied. One saved point was recorded after each gulp, and this process continued until the bottle was emptied. After that, the empty bottle was weighed and noted for comparison. For subsequent analysis and uncertainty propagation, all saved-point statistics and computed outputs from the full-bottle, intermediate-gulp, and empty-bottle conditions were kept. [2], [8]

III. RESULTS

Data was collected using the final DAQ sample windows, which were ± 2.56 V on ACH0, ± 0.64 V on ACH1, ± 1.28 V on ACH2, and ± 0.64 V on ACH3. In the analysis described, only ACH0 (V_s) and ACH2 (V_{amp}) were utilized.

Table I lists the measured beam dimensions and geometric input parameters utilized in the calculations for Mechanics of Materials. For all following MoM weight computations, these measured values were input into LabVIEW VI.

TABLE I
BEAM DIMENSIONS

Symbol	Value	Unit	Tool
b	0.02537 ± 0.0005	m	Caliper
h	0.00306 ± 0.00005	m	Micrometer
L	0.25 ± 0.005	m	Ruler
L_0	0.30 ± 0.005	m	Ruler

Table II reports the calibration measurements. Four applied calibration loads and one unloaded point were recorded. After linearizing the measured ΔV_{amp} and corresponding reference weights, a least-squares regression was carried out using the calibration data. The calibration constant, which came out to be 11.074 N/V, was determined by taking the slope of the best-fit line.

The Appendix contains the calibration plot used to determine c_{cal} . The uncertainty in c_{cal} was determined using Monte Carlo simulation, while propagated uncertainties in the calculated quantities were determined using root-sum-square (RSS) methods.

TABLE II
CALIBRATION WEIGHT MEASUREMENTS

ΔV_{amp} (V)	Weight (N)
0.000107 ± 0.0110	0.00000 ± 0.0005
0.0175 ± 0.0108	0.02000 ± 0.0005
0.0427 ± 0.0110	0.04905 ± 0.0005
0.0847 ± 0.0118	0.10000 ± 0.0005
0.177 ± 0.0134	0.19598 ± 0.0005

Table III lists the repeatability measurements for bottle positioning. The unopened bottle was placed ten times, and each saved point recorded the calibration-based weight estimate W_{cal} , the associated MoM weight estimate W_{MoM} , and the measured ΔV_{amp} . The ensuing statistical and uncertainty computations made use of these values.

TABLE III
REPEATABILITY MEASUREMENTS

ΔV_{amp} (V)	W_{MoM} (N)	W_{cal} (N)
0.215 ± 0.0106	2.02 ± 0.241	2.39 ± 0.239
0.214 ± 0.0094	1.99 ± 0.234	2.36 ± 0.230
0.220 ± 0.0090	2.04 ± 0.239	2.44 ± 0.234
0.215 ± 0.0150	2.00 ± 0.259	2.38 ± 0.264
0.222 ± 0.0129	2.07 ± 0.256	2.46 ± 0.257
0.219 ± 0.0093	2.04 ± 0.239	2.42 ± 0.234
0.206 ± 0.0211	1.92 ± 0.287	2.28 ± 0.306
0.223 ± 0.0100	2.08 ± 0.245	2.47 ± 0.241
0.221 ± 0.0093	2.06 ± 0.241	2.45 ± 0.236
0.227 ± 0.0100	2.12 ± 0.249	2.51 ± 0.245

Table IV provides a summary of the drinking measurements. Every row represents a single stored point that was gathered throughout the drinking process, including the calibration-based weight estimate W_{cal} , the MoM weight estimate W_{MoM} , and the observed ΔV_{amp} . The empty-bottle condition is represented by the final row in Table IV.

TABLE IV
DRINKING "GULP" MEASUREMENTS

ΔV_{amp} (V)	W_{MoM} (N)	W_{cal} (N)
0.220 ± 0.00909	2.05 ± 0.240	2.44 ± 0.235
0.188 ± 0.0140	1.75 ± 0.231	2.07 ± 0.237
0.141 ± 0.0168	1.31 ± 0.212	1.56 ± 0.230
0.0947 ± 0.00866	0.882 ± 0.125	1.05 ± 0.132
0.0608 ± 0.00904	0.567 ± 0.104	0.673 ± 0.116
0.0267 ± 0.0114	0.248 ± 0.110	0.295 ± 0.129
0.00640 ± 0.00861	0.0600 ± 0.0805	0.0708 ± 0.0956

For external comparison, the full bottle was measured on the laboratory scale as 2.4444 N, and the empty bottle was

measured as 0.0789 N. Using the drinking measurements in Table IV, the average gulp size was calculated as 0.395 N using the calibration method and 0.332 N using the Mechanics of Materials method. These values were used later in the comparison of the beam-scale methods with the direct scale measurements and in the evaluation of average gulp size.

IV. DISCUSSION

This experiment showed that both a calibration-based method and a Mechanics of Materials (MoM) method may be used to estimate bottle weight using the cantilever beam scale. Although the anticipated weight loss during the drinking sequence was recorded by both methods, the calibration method consistently yielded numbers that were nearer the direct laboratory scale measurements. In contrast to the laboratory scale value of 2.4444 N, the calibration approach yielded 2.4408 N, and the MoM method yielded 2.0536 N for the full-bottle scenario. In contrast to the laboratory scale value of 0.0789 N, the calibration and MoM techniques yielded 0.0708 N and 0.0596 N, respectively, for the empty-bottle scenario. These comparisons show that over the measured range, the calibrating approach produced the more accurate estimate.

For Hooke's law and the beam-bending relationships to continue to be applicable, the beam must stay within the linear elastic range for the MoM approach to be valid. The maximum allowed beam load was determined to be $43.7 \text{ N} \pm 4.76 \text{ N}$ using the 6061-T6 aluminum yield strength of $276 \text{ MPa} \pm 27.6 \text{ MPa}$. Compared to any load imposed during the experiment, this limiting load is far greater. The largest bottle weight recorded was 2.4444 N, a significant decrease from 43.7 N. As a result, it was anticipated that the beam would not give during testing, and all recorded data satisfied the elastic assumptions utilized in the MoM analysis.

The best indicator of which approach worked best in this configuration is the full-bottle comparison. The MoM estimate significantly underestimated the bottle weight, while the calibration-based estimate just slightly deviated from the laboratory scale result. This implies that the calibration technique more accurately captured the experimental system's real behavior. The calibration approach implicitly includes nonideal effects like the real amplifier response, slight variations in the effective load position, and other setup-specific characteristics because it was derived directly from measured voltage response vs applied weight. On the other hand, the MoM approach relied on the idealized beam model, measured geometry, and material properties; hence, any inaccuracy in those inputs had a direct impact on the weight prediction.

The measurement quality of the setup is also explained by the bottle-placement repeatability data. When the unopened bottle was placed repeatedly, the range of ΔV_{amp} values was small, suggesting that the bottle could be placed at the designated spot consistently. The spread in W_{MoM} and W_{cal} that resulted from the repeated measurements' differences, however, indicates that bottle positioning still introduced quantifiable variability. The calibration method remained closer to the direct scale measurement of the unopened bottle, as seen by the average repeatability values of 2.4183 N for the calibration method and

2.0349 N for the MoM method. The bottle-placement repeatability measurements were also treated as direct repeated measurements. Using a 95% confidence interval based on the t-distribution, the repeatability mean was $2.0349 \pm 0.0398 \text{ N}$ for the MoM method and $2.4183 \pm 0.0473 \text{ N}$ for the calibration method. These intervals show that repeated placement introduced measurable variability, while the calibration-based repeatability remained closer to the direct laboratory scale measurement of the unopened bottle. Even when the container was purposefully aligned to the same designated point, these repeated measurements verify that placement variation was one cause of uncertainty.

The drinking data provides a second way to compare the two methods. Both methods showed a monotonic decrease in estimated weight as successive gulps were taken, which is physically consistent with a decreasing bottle mass. Using the values from Table IV, the average gulp size was 0.395 N for the calibration method and 0.332 N for the MoM method. The average gulp sizes were also treated as direct measurements and analyzed using a 95% confidence interval based on the t-distribution. The resulting average gulp size was $0.332 \pm 0.096 \text{ N}$ for the MoM method and $0.395 \pm 0.114 \text{ N}$ for the calibration method. These intervals quantify the variability in gulp size across the drinking sequence and provide a statistical basis for comparing the two methods. Thus, both methods captured the same overall trend, but the calibration method predicted a larger average gulp size because it consistently reported larger bottle weights throughout the drinking sequence. This difference follows the same pattern already observed in the full-bottle repeatability measurements.

The measurement system's significant drawback is highlighted by the empty-bottle circumstance. Any baseline offset, amplifier drift, or DAQ resolution restriction had a greater relative impact on the computed weight at low loads since the measured ΔV_{amp} became small. Because of this, even though the nominal weights are minimal, the last rows of Table IV are nevertheless linked to significant uncertainty. Put otherwise, the usable signal got closer to the same scale as the zero-offset and noise contributions as the bottle got closer to the empty condition. Although both approaches became increasingly sensitive to minute voltage-level mistakes close to zero load, the calibration method stayed closer to the direct scale measurement at the empty-bottle state.

Since the calibration approach incorporated many of the actual experimental effects that the ideal beam model is unable to fully capture, it performed better than the MoM method. Even slight variations in b , h , L , and the expected material behavior have a direct impact on the MoM calculation, and they spread through the beam equation. Furthermore, the real bottle has a finite contact breadth, modest placement variation, and potential changes in the load distribution on the beam, whereas the model interprets the bottle as a concentrated load acting through a single effective center of gravity. The calibration method, by contrast, was built directly from measured system response, so it naturally incorporated the actual bridge/amplifier behavior and the effective sensitivity of the beam scale. For this reason, the calibration method was better

matched to the real apparatus used in the lab.

Several limitations of the setup suggest clear improvements for future testing. One important limitation was sensitivity to baseline drift in the amplified voltage V_{amp} . Although the bridge was balanced and the system was tared before testing, small changes in amplifier offset, wire position, strain-gage response, or thermal conditions could shift the unloaded baseline over time. Because all calculated loads depend on ΔV_{amp} , this zero-shift directly affects the final weights. Retaring before each new bottle measurement would improve accuracy by re-establishing the unloaded baseline immediately before each saved point. Additional improvements would include using a more controlled bottle-placement fixture, increasing the precision of geometric measurements, and collecting more calibration points over the operating range. Together, these changes would reduce both model-input uncertainty and voltage-offset uncertainty, which would improve the accuracy of the beam-scale measurements.

V. CONCLUSION

This experiment demonstrated that an instrumented cantilever beam can be used to estimate bottle weight and average gulp size using both a Mechanics of Materials (MoM) method and a calibration-based method. Both approaches captured the expected decrease in bottle weight during the drinking sequence, but the calibration method provided the most accurate overall results, giving 2.4408 N for the full bottle and 0.0708 N for the empty bottle compared with direct scale measurements of 2.4444 N and 0.0789 N, respectively. The average gulp size was found to be 0.395 N using the calibration method and 0.332 N using the MoM method. Because the maximum allowable beam load was 43.7 N, which was far greater than the loads applied during testing, the beam remained in the elastic range and the underlying model assumptions were satisfied. These results show that a strain-gage-based beam scale can provide reliable weight estimates when paired with appropriate calibration and uncertainty analysis, which is important for applications involving low-cost force measurement and experimental instrumentation. Future improvements should focus on reducing baseline drift by retaring before each new bottle measurement, improving placement consistency, and collecting additional calibration data to further reduce uncertainty and improve measurement accuracy.

APPENDIX

This appendix offers supporting calculations for the final reported results as well as a summary of the uncertainty methodologies employed in the current work. This includes root-sum-square (RSS) propagation for the final estimated numbers, Monte Carlo estimation of the calibration-constant uncertainty, independent input uncertainties, and statistical uncertainties from observed data. When applicable, the uncertainty treatment adheres to the definitions presented in the main report as well as the course uncertainty guidelines. [8]

A. Uncertainty of Given Values

Measured geometry, instrument resolution, course-given values, and published material properties were the sources of the independent uncertainties employed in this investigation. The propagated uncertainty calculations, which are detailed further in this appendix, were performed using these uncertainties as input values. While the material-property uncertainties for 6061-T6 aluminum were derived from the published reference used in the main paper, the geometric measurement uncertainties were based on the measurement instruments used during the experiment. [2], [7]

TABLE V
UNCERTAINTIES FOR GIVEN AND DIRECT MEASUREMENTS

Uncertainty in beam length	μ_{L_0}	± 0.005	m	Given [2]
Uncertainty of Length from Strain Gauge to Bottle Center of Mass	μ_L	± 0.005	m	Given [2]
Uncertainty of Beam Width	μ_b	± 0.005	m	Given [2]
Uncertainty of Beam Thickness	μ_h	± 0.00005	m	Instrument [2]
Uncertainty of Gain	μ_G	± 4.5	Unit less	Given [1]
Uncertainty of Gauge Factor	μ_{Gf}	± 0.01	Unit less	Given [1]
Uncertainty of Lab Scale	μ_{Ws}	± 0.0005	N	Given [2]
Uncertainty of Young's Modulus	μ_E	± 6.9	Gpa	Researched [7]
Uncertainty of Yield Stress	$\mu_{\sigma y}$	± 27.6	Mpa	Researched [7]

B. Statistical Uncertainty of Measured Data

Instrument and given-value uncertainties were handled separately from statistical uncertainty. The t-distribution was used to calculate the sample mean, sample standard deviation, standard error of the mean, and 95% confidence interval for measured data sets that were handled as direct measurements. This method was used for the average gulp-size computations and the bottle-placement repeatability data. In accordance with the course uncertainty guidance, the mean and standard deviation of 2500 samples were used directly as the statistical foundation for uncertainty calculations for the saved-point voltage measurements captured by the VI. [8]

$$\bar{x} = \frac{1}{N} \cdot \sum_{i=1}^N x_i \quad (17)$$

$$S_x = \sqrt{\frac{1}{N-1} \cdot \sum_{i=1}^N (x_i - \bar{x})^2} \quad (18)$$

$$S_{\bar{x}} = \frac{S_x}{\sqrt{N}} \quad (19)$$

$$P_x = t S_{\bar{x}} \quad (20)$$

TABLE VI
STATISTICAL UNCERTAINTIES FOR SAVED POINTS AND DIRECT MEASUREMENTS

Uncertainty of Supply Voltage	μ_{V_s}	2500	± 0.000288	V
Uncertainty of Amplified Bridge Voltage	$\mu_{V_{amp}}$	2500	± 0.011	V
Uncertainty of Tared Amplified Bridge Voltage	$\mu_{V_{tare}}$	2500	± 0.00068	V
Uncertainty of Weight from placement MoM	$\mu_{W_{Mp}}$	10	± 0.0398	N
Uncertainty of weight from Placement Cal	$\mu_{W_{Calp}}$	10	± 0.0473	N
Uncertainty of gulp size MoM	μ_{G_M}	6	± 0.096	N
Uncertainty of Gulp Size Cal	$\mu_{G_{Cal}}$	6	± 0.114	N

C. Determination of Calibration Constant

The calibration constant c_{cal} was determined from the linearized relationship between measured ΔV_{amp} and reference weight. One unloaded point and four applied calibration loads were used, and a least-squares regression was performed on the calibration data. The slope of the best-fit line was taken as the calibration constant, giving 11.074 N/V. The calibration plot used to determine this value is included below.

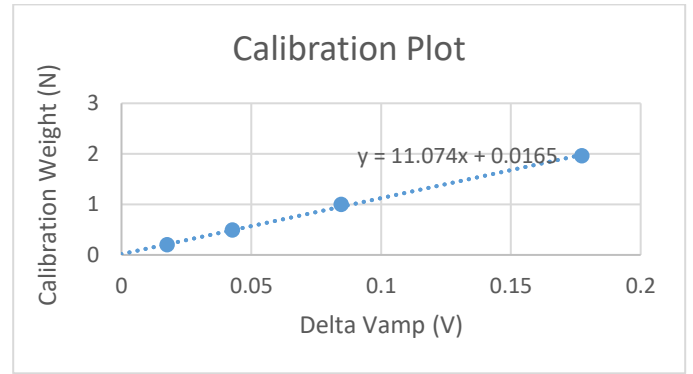


Fig 6. Calibration plot to determine calibration constant.

Uncertainty in c_{cal} was not determined using RSS propagation. Instead, a Monte Carlo simulation was used to estimate the uncertainty in the fitted slope based on the uncertainty in the calibration weights and the uncertainty in the measured voltage data. This approach was used because c_{cal} was obtained from a fitted regression rather than from a direct closed-form instrument equation. [8]

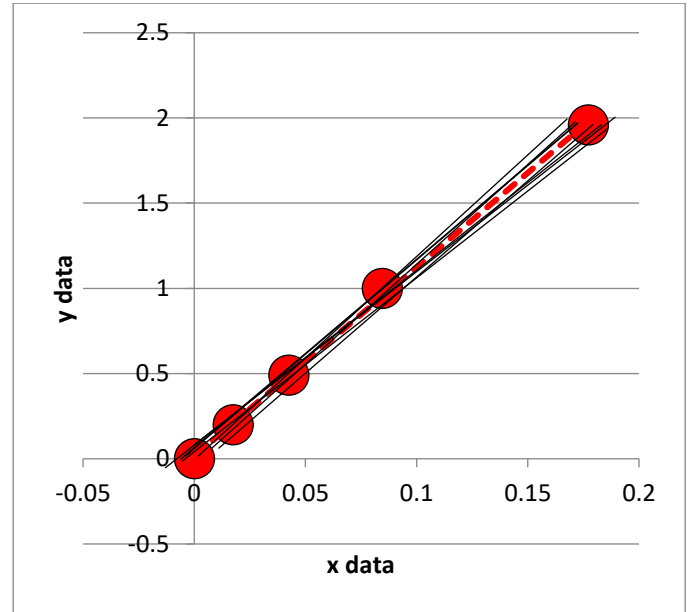


Fig 7. Monte Carlo simulation output used to estimate uncertainty for the calibration constant.

TABLE VII
MONTE CARLO CALIBRATION WEIGHTS AND UNCERTAINTIES

ΔV_{amp} (V)	Weight (N)
0.000107 ± 0.0110	0.00000 ± 0.0005
0.0175 ± 0.0108	0.02000 ± 0.0005
0.0427 ± 0.0110	0.04905 ± 0.0005
0.0847 ± 0.0118	0.10000 ± 0.0005
0.177 ± 0.0134	0.19598 ± 0.0005

D. RSS Propagation of Tared Voltage ΔV_{amp}

The tared amplified voltage ΔV_{amp} was obtained from the measured amplifier output and the unloaded baseline tare value.

The uncertainty in ΔV_{amp} was determined using root-sum-square propagation of the independent uncertainties associated with these measured voltage terms. The resulting propagated uncertainty in ΔV_{amp} was then used in both the calibration-based and MoM-based weight calculations.

$$\mu_{\Delta V_{amp}} = \left[\left(\mu_{V_{amp}} \frac{\partial \Delta V_{amp}}{\partial V_{amp}} \right)^2 + \left(\mu_{V_{tare}} \frac{\partial \Delta V_{amp}}{\partial V_{tare}} \right)^2 \right]^{\frac{1}{2}} \quad (21)$$

E. RSS Propagation of W_{MoM}

Rather than propagating uncertainty through a separate intermediate strain calculation, uncertainty in the Mechanics of Materials method was propagated directly through the combined W_{MoM} equation. This allowed the final uncertainty in the MoM weight estimate to be obtained in one step without introducing a separate strain-propagation section.

$$\mu_{W_{MoM}} = \left[\left(\mu_{\Delta V_{amp}} \frac{\partial W_{MoM}}{\partial \Delta V_{amp}} \right)^2 + \left(\mu_G \frac{\partial W_{MoM}}{\partial G} \right)^2 + \left(\mu_{V_s} \frac{\partial W_{MoM}}{\partial V_s} \right)^2 + \left(\mu_{G_f} \frac{\partial W_{MoM}}{\partial G_f} \right)^2 + \left(\mu_E \frac{\partial W_{MoM}}{\partial E} \right)^2 + \left(\mu_b \frac{\partial W_{MoM}}{\partial b} \right)^2 + \left(\mu_h \frac{\partial W_{MoM}}{\partial h} \right)^2 + \left(\mu_L \frac{\partial W_{MoM}}{\partial L} \right)^2 \right]^{\frac{1}{2}} \quad (22)$$

F. RSS Propagation of W_{cal}

The calibration-based weight was determined from the calibration constant and the measured tared voltage. Therefore, the uncertainty in W_{cal} was propagated from the uncertainty in c_{cal} and the uncertainty in ΔV_{amp} . The uncertainty in c_{cal} was taken from the Monte Carlo simulation described above, and the uncertainty in ΔV_{amp} was taken from the direct RSS propagation of the tared voltage.

$$\mu_{W_{cal}} = \left[\left(\mu_{\Delta V_{amp}} \frac{\partial W_{cal}}{\partial \Delta V_{amp}} \right)^2 + \left(\mu_{c_{cal}} \frac{\partial W_{cal}}{\partial c_{cal}} \right)^2 \right]^{\frac{1}{2}} \quad (23)$$

G. RSS Propagation of Maximum Allowable Load

The maximum allowable load before yielding was used in the Discussion to verify that the beam remained in the elastic range during testing. The uncertainty in W_{max} was propagated using the independent uncertainties in yield strength, beam width, beam thickness, and lever arm distance. This provided the uncertainty associated with the elastic-limit comparison used to justify the validity of the MoM assumptions.

$$\mu_{W_{max}} = \left[\left(\mu_{\sigma_y} \frac{\partial W_{max}}{\partial \sigma_y} \right)^2 + \left(\mu_b \frac{\partial W_{max}}{\partial b} \right)^2 + \left(\mu_h \frac{\partial W_{max}}{\partial h} \right)^2 + \left(\mu_L \frac{\partial W_{max}}{\partial L} \right)^2 \right]^{\frac{1}{2}} \quad (24)$$

H. Propagated Uncertainties of Repeatability and Gulp

The propagated uncertainties for the bottle-placement repeatability measurements and the drinking measurements were evaluated using the uncertainty relations described above. For each saved point, the reported uncertainty in W_{MoM} was obtained from the direct RSS propagation of the combined MoM equation, and the reported uncertainty in W_{cal} was obtained from the propagated uncertainty in the calibration method. These propagated uncertainties are summarized in the tables below.

TABLE VIII
REPEATABILITY UNCERTAINTIES

$\mu_{\Delta V_{amp}}$ (V)	$\mu_{W_{MoM}}$ (N)	$\mu_{W_{cal}}$ (N)
± 0.0105	± 0.241	± 0.238
± 0.00944	± 0.234	± 0.230
± 0.00902	± 0.238	± 0.233
± 0.0149	± 0.258	± 0.264
± 0.0129	± 0.256	± 0.257
± 0.00928	± 0.238	± 0.234
± 0.0210	± 0.287	± 0.306
± 0.0100	± 0.245	± 0.241
± 0.00933	± 0.240	± 0.236
± 0.00998	± 0.249	± 0.244

TABLE IX
DRINKING “GULP” UNCERTAINTIES

$\mu_{\Delta V_{amp}}$ (V)	$\mu_{W_{MoM}}$ (N)	$\mu_{W_{cal}}$ (N)
± 0.00909	± 0.239	± 0.234
± 0.0139	± 0.231	± 0.237
± 0.0168	± 0.212	± 0.230
± 0.00868	± 0.125	± 0.132
± 0.00904	± 0.104	± 0.115
± 0.0114	± 0.110	± 0.129
± 0.00861	± 0.0805	± 0.0955

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